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High incidence of Type 2 diabetes mellitus in South African Indians: a 10-year follow-up study

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Abstract

Aims Previous cross-sectional studies have established that South African Indians have a high prevalence of Type 2 diabetes mellitus. A prospective community study was undertaken to determine the incidence of Type 2 diabetes and the risk factors associated with its development in a cohort of South African Indians who had been studied 10 years previously.

Methods This is a report on 563 subjects who participated both at baseline and at the 10-year follow-up study. In the baseline study, 2479 subjects (> 15 years) were studied; using 1985 World Health Organization criteria for glucose tolerance based on 75 g oral glucose tolerance tests (OGTT), the crude prevalence of diabetes mellitus (Diabetes) was 9.8% and of impaired glucose tolerance (IGT) 5.8% (age and sex-adjusted prevalence 13% and 6.9%, respectively).

Results At the 10-year follow-up study, 563 of the subjects who could be traced consented to a repeat OGTT; of these, 91 (16.2%) were classified as Diabetes and 41 (7.3%) as IGT. Of the subjects who did not have diabetes at baseline (n = 517), 49 (9.5%) progressed to diabetes (PTD) and 40 (7.7%) had IGT. The crude cumulative incidence of diabetes was 9.5% (rate of progression 0.95% per annum; incidence density 9.5/1000 person years) with an age and sex-adjusted cumulative incidence of 8.3% (rate of progression 0.95% per annum; incidence density 8.3/1000 person years). Examination of risk factors predictive of subsequent diabetes development was undertaken by analysis of baseline (year 0) variables in the 517 subjects who did not have diabetes at baseline. In multivariate analysis using a logistic regression model, the significant predictive risk factors for future diabetes included 2-h post load plasma glucose (2 PG) ($P < 0.0001$, odds ratio (OR) 1.7, 95% confidence interval (CI) 1.4–2.1), body mass index (BMI) ($P < 0.006$, OR 1.1, 95% CI 1.0–1.3) and obesity ($P < 0.01$, OR 4.6, 95% CI 1.4–14.7).

Conclusions This long-term study has shown that in South African Indians there is a high incidence of Type 2 diabetes, and in this population significant predictors include higher baseline blood glucose, BMI and obesity.

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Keywords South African Indians, incidence, risk factors, obesity, body mass index, plasma glucose

Abbreviations WHO, World Health Organization; IGT, impaired glucose tolerance; 2 PG, 2-h post load plasma glucose; FPG, fasting plasma glucose

Introduction

After the introduction of standardized criteria for glucose tolerance by the World Health Organization (WHO) [1,2] and National Diabetes Data Group [3], there has been considerable

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information on the prevalence of Type 2 diabetes and impaired glucose tolerance (IGT) from cross-sectional studies in various populations [4,5]. However, data on long-term prospective population studies using WHO criteria are limited [6–12]. In recent years many prospective studies have focused attention either on populations with a high prevalence of diabetes [6–11] or on defined high-risk groups, e.g. IGT [13–28] or fasting hyperglycaemia [29].

From population studies, the crude incidence of diabetes varies, from 1.2 to 30.5/1000 person years (p.y.) [6–12]. A low incidence (< 5/1000 p.y.), has been reported in Papua New Guinea Highlanders (Melanesians), Europid Americans and Frenchmen [6,11,25], intermediate rates (5–15/1000 p.y.) in Creole and Chinese Mauritians, Maltese, Mexican Americans and rural Samoans [6,10–12], while the highest incidence (> 15/1000 p.y.) has been observed in Pima Indians, rural Wanigelas in Papua New Guinea, Nauruans, urban Samoans and Mauritian Indians [6–9]. However, comparison between studies is hampered by differences in age structure and levels of obesity in the different study populations [6]. Higher rates of progression to Type 2 diabetes have been reported in subjects who had IGT at baseline [6–13,22,24–29].

Several risk factors for the development of diabetes have been identified. In subjects who are non-diabetic (IGT + normal glucose tolerance (NGT)) at baseline, the most consistent significant independent risk factor is the blood glucose concentration [8–10,22,30], both fasting [8,10,30] and 2-h post load plasma glucose [8,9,22,30]; other less consistent risk factors include body mass index (BMI) [8,11,22,31], serum insulin [8–10,30,32], family history [30,31], age [9,11,22,30,31] and gender [9].

A previous cross-sectional population study has established that South African Indians, a migrant Asian group, have a high prevalence of diabetes [33,34]. In a 4-year prospective study of subjects with IGT in this population, a strikingly high risk of progression to diabetes was observed [26]. However, there are no published reports of long-term prospective studies in subjects with normal glucose tolerance or IGT in this population.

This study was undertaken to evaluate the glucose tolerance status of South African Indians who had originally been examined 10 years previously, to determine the incidence of diabetes and the risk factors associated with its development.

Patients and methods

Background

South African Indians, whose ancestors migrated from the Indian subcontinent during the latter half of the 19th century, comprise two ethnic subgroups, the Aryans (Hindi, Gujerathi and Urdu speaking) and Dravidians (Tamil and Telegu speaking), from Northern and Southern India, respectively; two-thirds are South Indians and one-third North Indians [26,34,35]. The present population of just over 1 million have all been born in South Africa, and almost 60% live in seven residential areas in the city of Durban. Prior to political independence in 1994, by virtue of the apartheid system and its Group Areas Act, different ethnic groups were forced by legislation to live in separate residential areas; in 1991, these laws were repealed and there ensued major inter- and intra-suburban and city relocations.

Baseline study (year 0)

A detailed description of the baseline study has been previously published [33,34]. In brief, in 1984 (year 0 of the study), a cross-sectional epidemiology survey was undertaken to determine the prevalence of NIDDM among South African Indians living in seven residential areas in Durban. The study was conducted by systematic cluster sampling of households on all non-pregnant adult family members > 15 years of age, including subjects known to have diabetes. In all consenting eligible subjects, a modified 75 g oral glucose tolerance test (OGTT) was performed, using WHO criteria for field conditions [1,2]. The positive response rate was 92%. Of the 2479 (1441 women; 1038 men), the crude prevalence of diabetes mellitus was 9.8% and of IGT 5.8%; the age and sex-adjusted prevalence was 13% and 6.9%, respectively.

Follow-up study (year 10)

In 1994 (year 10 of the study), a follow-up study was undertaken to re-evaluate all subjects who were studied at baseline. The recruitment process was as for the baseline study. Both positive and negative responses were recorded. (A non-responder was defined as one who was contacted on at least three occasions.)

Subjects were classified as 'traced' or 'not traced' (not contactable). The subjects who could be traced were further classified as 're-test' (consented to re-evaluation), 'known deaths' and 'refused'; of those who were eligible for re-evaluation, a 'refusal/non-responder' was one who had been contacted but refused on at least three occasions.

Informed consent was obtained from all participants and the University of Natal Ethics Committee approved the study.

Subjects

This is a report on 563 subjects (331 women and 232 men) who were studied both at baseline (year 0) and at the 10-year follow-up study.

Survey procedure

In the baseline study, following an overnight fast, a modified 75 g OGTT was performed [1,2]; height and weight were measured for determination of BMI, and details of age, gender, language, religion, personal and family history of diabetes were recorded. A subset of subjects had blood pressure examination and assessment of serum insulin during OGTT [26,33,34].

In the follow-up study, the additional variables which were examined on all subjects included waist and hip circumference for calculation of waist–hip ratio (WHR), blood pressure, fasting serum total cholesterol and triglycerides, serum uric acid and fasting and 2-hr post load serum total insulin, intact insulin, proinsulin and conversion intermediates; a subset had

HbA_{1c} levels assessed and DNA for candidate gene analysis. This report is confined to those variables examined both at baseline and in the follow-up study.

BMI was calculated as (weight in kg) ((height in metres)^2; obesity was defined as BMI (kg/m^2) (27 in men and 25 in women) [26,33,34,36]. Waist circumference was measured at the level of the umbilicus and hip circumference at the level of the greater trochanter. The WHR was used as a measure of upper body obesity.

Risk factor analysis

Examination of risk factors predictive of progression to Type 2 diabetes (PTD) was undertaken in those subjects who did not have diabetes at baseline (NGT + IGT). Baseline (year 0) variables (n = 35) analysed included age, gender, reported family history of diabetes, BMI, and fasting (FPG) and 2-h post load (2 PG) plasma glucose during OGTT. In a subset of subjects (n = 283), systolic and diastolic blood pressure was analysed.

OGTT and classification of glucose tolerance

A modified 75 g OGTT [2] was performed, i.e. venous plasma glucose after an overnight fast and 2 h after ingestion of 75 g of glucose monohydrate dissolved in 250 ml of water. Plasma glucose was measured by a glucose oxidase method using an autoanalyser [26,34].

Glucose tolerance classification was based on 1985 WHO criteria [2]. IGT was defined by FPG < 7.8 mmol/l and 2 PG ≥ 7.8 but < 11.1 mmol/l; diabetes mellitus (Diabetes) was diagnosed if FPG ≥ 7.8 mmol/l and/or 2 PG ≥ 11.1 mmol/l; NGT was defined if both FPG and 2PG were < 7.8 mmol/l.

Statistical analysis

Statistical analyses were performed using the SAS statistical package [37]. Data are presented as mean ± SD. For numerical data, when more than two groups were being compared, analysis of variance (ANOVA) was used. Duncan's multiple range test was used to evaluate the difference between any two groups studied. Where only two groups were being compared, Student's t-test was used. A χ^2 test was used for categorical data.

In the univariate analysis, to assess which factors were associated with PTD, the means of selected clinical and biochemical variables were compared in the group that progressed to diabetes and the group that did not progress; comparison was performed by Student's t-test or χ^2 method.

In the multivariate analysis, to evaluate the effect of various predictor variables, the multiple logistic regression model was used to predict progression to Type 2 diabetes. All baseline variables significant in univariate analysis were entered into the model. Variables not meeting the criteria for significance were then excluded in a stepwise fashion with the use of a backward elimination method [38]. The magnitude of the risk ratio is presented as the odds ratio (OR) with 95% confidence interval (CI) and P-value.

The age-standardized incidence was calculated with the direct method of adjustment using the world population as the standard [39].

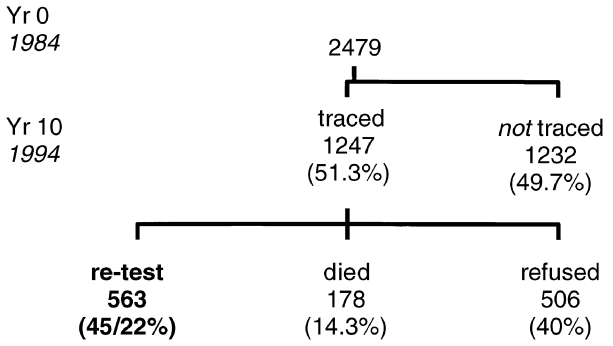

Figure 1 Flowchart of the eligibility and response in the 10-year follow-up study.

Table 1 Glucose tolerance status in 563 subjects who were tested both at baseline and the 10-year follow-up study

		Glucose tolerance		
		10-year follow-up, n (%)		
Baseline	n (%)	Normal	IGT	Diabetes
Normal	482 (85.6)	412 (85.5)	34 (7.1)	36 (7.5)
IGT	35 (6.2)	16 (45.7)	6 (17.1)	13 (37.1)
Diabetes	46 (8.2)	3 (6.5)	1 (2.2)	42 (91.3)
Normal + IGT	517 (91.8)	428 (82.8)	40 (7.7)	49 (9.5)
Total	563 (100)	431 (76.6)	41 (7.3)	91 (16.2)

Results

Follow-up glucose tolerance status

Response rate

Of the original eligible cohort of 2479 subjects, 1247 (51.3%) were traced; of these, 506 (40.6%) refused re-evaluation, 178 (14.3%) were known to have died and 563 (45.2% of 'traced' subjects; 22.7% of original cohort) consented to re-evaluation ('re-test') (Fig. 1).

Follow-up OGTT results (Table 1)

At the 10-year follow-up study, of the 563 eligible subjects who consented to a repeat OGTT, 431 (76.6%) had NGT, 41 (7.3%) had IGT and 91 (16.2%) had diabetes.

Of the 517 subjects who did not have diabetes at baseline (i.e. NGT + IGT), 49 (9.5%) developed diabetes (PTD) at follow-up, 40 (7.8%) had IGT and 428 (82.8%) had NGT. The crude cumulative incidence of diabetes was 9.5% (9.5/1000 person years; rate of progression 0.95% per annum); the age and sex-adjusted incidence was 8.3% (8.3/1000 person years; rate of progression 0.83% annum).

Of the 482 subjects who had NGT at baseline 36 (7.5%) had developed diabetes (PTD) and 34 (7.1%) progressed to IGT at follow-up.

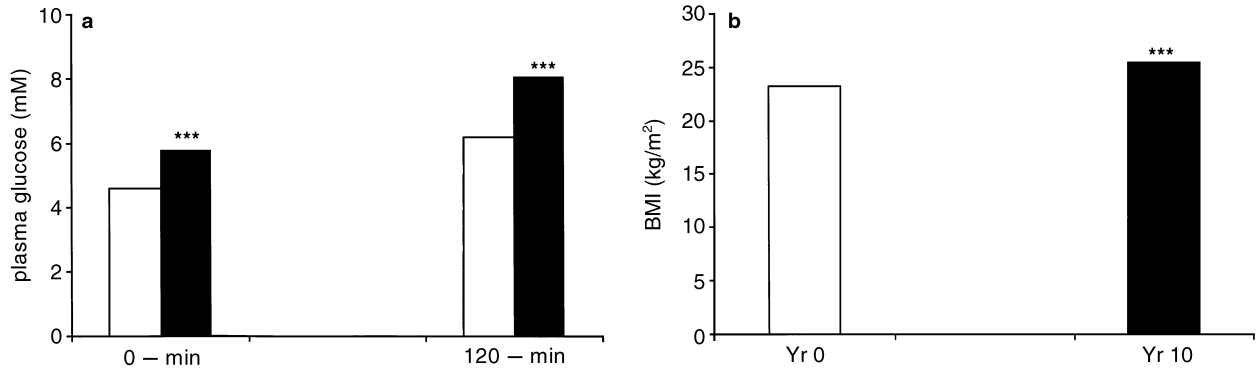

Figure 2 Mean plasma glucose (a) and body mass index (BMI) (b) at baseline (year 0) (□) and at follow-up (year 10) (■) in the total group (n = 563); ***P < 0.001, year 10 vs. year 0.

Table 2 Clinical and biochemical characteristics at baseline (year 0) in responders (n = 563) and non-responders (n = 1916)

	Responders	Non-responders	P
n	563	1916	
Sex (M/F)	226/337	806/1110	NS
Age (years)	36.4 ± 13.9	34.1 ± 15.9	0.0009
Family history (%)			
Father	16.4	14.7	NS
Mother	28.8	26.7	0.02
Brothers	8.7	6.6	NS
Sisters	13.6	9.7	NS
Sons	0.6	0.9	NS
Daughters	0.9	1.5	NS
BMI (kg/m^2)	22.6 ± 6.0	20.6 ± 8.6	0.0001
Obese (%)	26.8	23.0	NS
Blood pressure (mmHg)			
Systolic	118.9 ± 18.8	119.2 ± 22.7	NS
Diastolic	77.8 ± 12.9	77.8 ± 14.9	NS
Plasma glucose (mmol/l)			
0 min	4.6 ± 1.8	4.8 ± 2.6	0.043
120 min	6.2 ± 3.8	6.7 ± 5.2	0.017

Data are means ± SD except as noted.

Change from baseline

In the group as a whole (n = 563), when compared with baseline (year 0), there was a significant increase at 10-year follow-up study for the following variables: fasting plasma glucose (4.6 + 1.9 vs. 5.8 + 2.6 mmol/l, P < 0.001); 2-h plasma glucose (6.2 + 3.7 vs. 8.1 + 5.4 mmol/l, P < 0.001); BMI (23.2 + 5.0 vs. 25.5 + 5.3 kg/m^2, P < 0.001) (Fig. 2), and prevalence of obesity (27.2% vs. 43.1%, P < 0.001).

Responders vs. non-responders (Table 2)

At baseline (year 0), when compared with non-responders, the responders as a group were older (P < 0.001), heavier (P < 0.001), had lower FPG (P < 0.05) and 2 PG (P < 0.05), and a higher prevalence of a history of maternal diabetes. Of the 178 subjects who died, 30.3% had Diabetes at baseline and 9.0% had IGT.

Risk factors associated with diabetes development

The following outlines the analysis in 517 subjects who did not have diabetes at baseline.

Univariate analysis

Table 3 shows the baseline (year 0) clinical and biochemical characteristics, based on category of glucose tolerance and in relation to PTD, at follow-up (year 10). The mean age (P < 0.001), weight (P < 0.001), BMI (P < 0.001), FPG (P < 0.001) and 2 PG (P < 0.001) were significantly higher in the group which progressed to diabetes (PTD) than in the group that did not (not PTD).

The proportion of subjects who progressed to diabetes (PTD) increased with increasing baseline 2-h plasma glucose and BMI. Both of the subjects who at baseline had 2 PG > 10.3 to < 11.1, progressed to diabetes at follow-up (Fig. 3).

The mean BMI and prevalence of obesity was significantly higher at follow-up than at baseline, both in the group that PTD and in the group that did not PTD; however, the average difference was similar between the two groups (Table 4).

Multivariate analysis

Multivariate analysis showed that the significant predictive variables for subsequent diabetes development in the final model included 2-h plasma glucose (P < 0.0001, OR 1.7, 95% CI 1.4–2.1), BMI (P < 0.006, OR 1.1, 95% CI 1.0–1.3) and obesity (P < 0.01, OR 4.6, 95% CI 1.4–14.7) (Table 5).

Conclusions

This 10-year prospective population study has shown that in South African Indians there is an increased incidence of Type 2 diabetes mellitus, and that the significant predictors for its development include a higher 2-h post load plasma glucose, BMI and obesity at baseline.

Although there is considerable evidence that the prevalence of Type 2 diabetes and IGT varies between populations [4,5], there have been fewer comparative data available for the incidence of NIDDM in different populations; comparison is

Table 3 Baseline (year 0) clinical and biochemical variables, based on the category of glucose tolerance and in relation to progression to Type 2 diabetes (PTD) at the 10-year follow-up study, in 517 subjects who had normal glucose tolerance ($n = 482$) or impaired glucose tolerance (IGT) ($n = 35$) at baseline

	Glucose tolerance at follow-up (year 10)					
	Normal	IGT	D (PTD)	Not PTD	*P**	*P***
n	428	40	49	468		
Sex (M/F)	173/255	19/21	23/26	192/276	NS	NS
Age (years)	34.2 ± 13.7	39.5 ± 14.6	42.0 ± 11.3†	34.6 ± 13.8	0.0001	0.0003
Family history (%)						
Father	13.4	16.2	12.8	13.6	NS	NS
Mother	26.4	37.8	30.4	27.4	NS	NS
Brothers	5.6	20.0	6.8	6.9	0.035	NS
Sisters	8.9	11.8	11.6	9.2	NS	NS
Sons	1.1	0.0	2.5	1.0	NS	NS
Daughters	0.0	0.0	0.0	0.0	NS	NS
BMI (kg/m^2)	22.5 ± 5.0‡	23.7 ± 4.7	25.2 ± 4.3	22.6 ± 5.0	0.0009	0.0005
Obese (%)	23.3	30.8	31.3	23.8	NS	NS
Blood pressure (mmHg)§						
Systolic	118.3 ± 7.1	116.3 ± 3.8	122.3 ± 19.4	118.1 ± 16.8	NS	NS
Diastolic	77.0 ± 11.3	76.8 ± 10.9	80.1 ± 11.8	77.0 ± 11.2	NS	NS
Plasma glucose (mmol/l)						
0 min	4.1 ± 0.8¶	4.3 ± 0.9	4.7 ± 1.2	4.1 ± 0.8	0.0001	0.0009
120 min	5.1 ± 1.2	5.9 ± 1.6	6.6 ± 1.8††	5.1 ± 1.2	0.0001	0.0001

Data are means ± SD except as noted.
**P*-values for comparison between the Normal vs. IGT vs. Diabetes groups (analysis of variance or χ^2 method).
***P*-values for comparison between PTD vs. Not PTD (*t*-test or χ^2 method).
†Diabetes vs. Normal and IGT, significant difference using Duncan's multiple range test.
‡Normal vs. Diabetes, significant difference using Duncan's multiple range test.
§283 subjects only.
¶Normal vs. IGT and Diabetes, significant difference using Duncan's multiple range test.
††Normal vs. IGT vs. Diabetes, significant difference using Duncan's multiple range test.

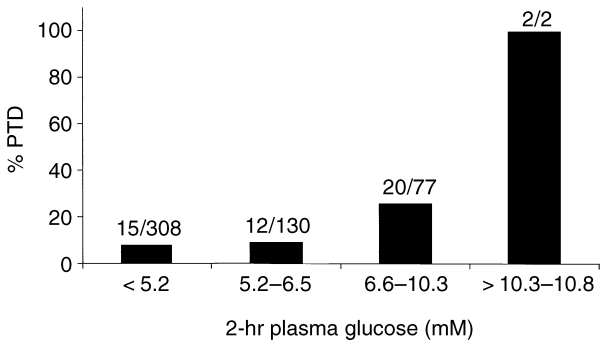

Figure 3 Proportion (%) of non-diabetic (normal glucose tolerance (NGT) + impaired glucose tolerance (IGT)) subjects who progressed to diabetes (PTD) at the 10-year follow-up, in relation to baseline (year 0) 2-h plasma glucose concentration. Above each bar, the numerator represents the number of subjects who progressed to diabetes, and the denominator represents the total number in that quartile.

Table 4 Body mass index (BMI) and prevalence of obesity at baseline and at the 10-year follow-up in the group which progressed to NIDDM (PTD) ($n = 49$) and the group which did not progress (not PTD) ($n = 468$)

	Year 0	Year 10	*P*
BMI (kg/m^2)*			
PTD ($n = 49$)	25.2 + 4.3	27.1 + 4.7	< 0.01
Not PTD ($n = 468$)	22.6 + 5.0	25.3 + 5.4	< 0.001
P	< 0.001	< 0.05	
Obese (%)			
PTD ($n = 49$)	31.9	53.2	< 0.05
Not PTD ($n = 468$)	23.9	42.5	< 0.001

*Average BMI difference (year 10 – year 0) in PTD (1.8 ± 3.9) vs. not PTD (2.7 ± 6.1), *P* = NS.

further hampered by the differences in age structure, duration and levels of obesity in the different study populations [6].

Comparison with other population studies which are based on 1985 WHO criteria and which use crude incidence rates [6–12,25,40] has shown that in South African Indians, the incidence of diabetes (9.5/1000 p.y.) is lower than in such high-risk populations as Pima Indians (30.5/1000 p.y.) and Pacific Islanders (Wanigelas 24/1000 p.y.; Nauruans 22.5/1000 p.y.; urban Samoans (16.6/1000 p.y.) and Mauritians (Indians 15.8/1000 p.y.; Creoles 12.2/1000 p.y., Chinese 10.4/1000 p.y.) [6–8]. When compared with Mexican Americans (8.1/1000 p.y.) [11], another high-risk population, the incidence was higher in South African Indians; the rate was certainly higher than in known low-risk populations, e.g. Papua New Guinea Highlanders (1.2/1000 p.y.), non-Hispanic White Americans (2.5/1000 p.y.), Frenchmen (4.2/1000 p.y.) and rural Samoans (5.7/1000 p.y.) [6,11,25].

A major limitation of this study was the low response rate (45.2% of traced cohort and 22.7% of original cohort).

Table 5 Multiple logistic regression predicting progression to NIDDM in 517 subjects who were non-diabetic (normal glucose tolerance (NGT) + impaired glucose tolerance (IGT)) at baseline

	PE	SE	χ^2	*P*	OR	95% CI
Initial step						
Age (years)	0.004	0.01	0.08	0.8	1.004	0.98–1.0
BMI (kg/m^2)	0.11	0.05	6.24	0.01	1.12	1.02–1.23
0 min plasma glucose (mmol/l)	−0.07	0.17	0.18	0.67	0.93	0.67–1.30
120-min plasma glucose(mmol/l)	0.44	0.09	26.07	0.0001	1.55	1.31–1.83
Obese (yes/no)	0.85	0.52	2.65	0.1	2.35	0.84–6.56
Final step						
BMI (kg/m^2)	0.13	0.05	7.72	0.006	1.14	1.04–1.26
120-min plasma glucose(mmol/l)	0.55	0.10	28.86	0.0001	1.74	1.42–2.13
Obese (yes/no)	1.52	0.59	6.63	0.01	4.59	1.44–4.65

PE, Parameter estimate; SE, standard error; χ^2, chi-square test; OR, odds ratio; CI, confidence interval.

Whether the incidence was overestimated because the responders were older and heavier is difficult to answer, but this group had lower FPG and 2 PG at baseline. Attrition rates in long-term studies are well recognized; in this study, this was compounded by the recent socio-political change, which resulted in major intersuburban/city relocations; in the 4-year prospective study on IGT subjects drawn from the original cohort, the response rate was 89.5% [26]. Moreover, the response rate in this study was similar across the groups of glucose tolerance. Also, the crude incidence of diabetes (9.5%) in this study correlates with the crude prevalence (9.8%) reported at baseline [34].

BMI and obesity were found to be significant predictive risk factors for subsequent diabetes in this study; obese subjects had a 4.6-fold greater risk of progressing to diabetes than did non-obese subjects. When compared with other studies, which used similar methodologies, in subjects who were non-diabetic at baseline, BMI was shown to be a significant predictor in Pima Indians [8,31] and Nauruans [22], the two populations with the world's highest prevalence and incidence of NIDDM. In Mexican Americans, although BMI was significant in univariate analysis, it failed to emerge as a significant variable in multivariate analysis [10]; in a subsequent report, which included Europid Americans, BMI was found to be significant [11]. In the low-prevalence French population, BMI was a significant predictor of future diabetes in subjects who were either NGT or IGT at baseline [25]. Most prospective studies have focused on high-risk groups, e.g. IGT; many such studies, including the South African study, although showing higher baseline BMI and obesity in subjects who PTD [9,12,13,16,18–22,24–29], failed to show these as independent predictors in multivariate analysis [13,18,21,25,29]. This may be accounted for, as suggested by Keen [18], by the short duration of follow-up, and that if followed-up for longer periods, the significance of BMI may be highlighted. In subjects with transient IGT, obesity was found to be a risk factor for PTD in Pima Indians [41], while absence of obesity was a protective factor (predictor of reversion to NGT) in South African Indians [42]. This may concur with Saad's suggestion [43] that obesity exerts its main effect on deterioration from NGT to IGT, while β-cell dysfunction plays a role in progression of IGT to Diabetes. The observation that BMI and obesity are significant risk factors for PTD in South African Indians may have important prognostic implications, since obese subjects may benefit from lifestyle interventions aimed at reducing weight.

A higher 2-h plasma glucose at baseline emerged as a significant predictor of future diabetes; this is consistent with previous studies, which have established the significance of baseline plasma glucose as a predictor [8–10,12,15–22,26–28,30]. Of interest was that the baseline FPG, although significant in univariate analysis, failed to emerge as an independent predictor in multivariate analysis. From previous studies which have examined plasma glucose as a risk factor in non-diabetic subjects (IGT + NGT) at baseline, variable results have been reported [8–10,12,22,30]; 2-h plasma glucose alone was found to be an independent predictor in Nauruans [9,22], fasting but not 2-h levels in Mexican Americans [10] and both fasting and 2-h plasma glucose in Pima Indians [8,30]. By contrast, in most studies, which examined the natural history of IGT, both fasting and 2-h levels were significant predictors of future diabetes [13,15,16,18–22,24,26–28]. This concurs with Knowler *et al.*'s [8] suggestion that there is little relationship with 2-h plasma glucose in the range considered NGT by 1985 WHO criteria (< 7.8 mmol/l), but that individuals with higher 2-h levels (IGT range) have markedly higher incidence rates of diabetes.

As in other studies [8,25,30], using cut-off levels, the proportion of subjects progressing to diabetes increases with increasing baseline 2-h plasma glucose. It is the observation that both the subjects with baseline 2-h plasma glucose > 10.3 mmol/l subsequently progressed to diabetes; this cut-off level is remarkably similar to that found in the 4-year study in IGT subjects in this population, in which the risk was total if 2-h plasma glucose was > 10.3 mmol/l at baseline [26]. This may have prognostic implications regarding advice on lifestyle modifications to lower blood glucose in this population.

In conclusion, this long-term study has shown that in South African Indians, there is an increased incidence of Type 2 diabetes, and that significant predictive risk factors for its development include a higher 2-h post load plasma glucose, BMI and obesity at baseline.

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